R13

Code No: 115AN

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year I Semester Examinations, July/August - 2023
PRINCIPLES OF PROGRAMMING LANGUAGES
(Computer Science and Engineering)

Time: 3 hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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PART - A

(25 Marks)

1.a)	Compare the behavior of imperative paradigm and functional paradigm.	[2]
b)	What is virtual machine, specify its purpose?	[3]
c)	What is strong typing?	[2]
d)	What is a pointer variable? How does it differ with normal variable?	[3]
e)	What is static scope?	[2]
f)	Differentiate '&' with '&&'.	[3]
g)	List out the importance of Polymorphism.	[2]
h)	What is the importance of the Event handler?	[3]
i)	Write a code for subtraction of two numbers using imperative language.	[2]
j)	Write a program to find out the area of a triangle using functional paradigm.	[3]

PART - B

(50 Marks)

2. Discuss on:
a) The grammar for simple statements
b) Attribute Grammar and Ambiguous Grammar. [5+5]

OR

3.a) Derive the Parse tree for the statement A= A+(B*C).
b) Explain the need and importance of syntax and semantics. [6+4]

4. Explain about:
a) Conditional statements
b) Looping statements with neat sketch and example. [5+5]

OR

5. Explain the various expressions with suitable example. [10]

6.a) Explain scope and lifetime of a variable using examples.
b) Distinguish between static and dynamic scoping. [5+5]

OR

7. Write a Program to find the sum of two complex numbers using user defined overloading concepts. [10]

8.a) Narrate on the elements of prolog
b) What is the difference between the C++ event handler and JAVA event handler? [3+7]

OR

9.a) Explain about assertions in exceptions.
b) Enumerate on the behavior of finally keyword. [5+5]

10.a) List out the behavioral differences of procedural abstraction and data abstraction.
b) Explain about the values and types of python. [4+6]

OR

11.a) Write notes on first functional programming language.
b) Explain about the importance of bindings and scope. [5+5]

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